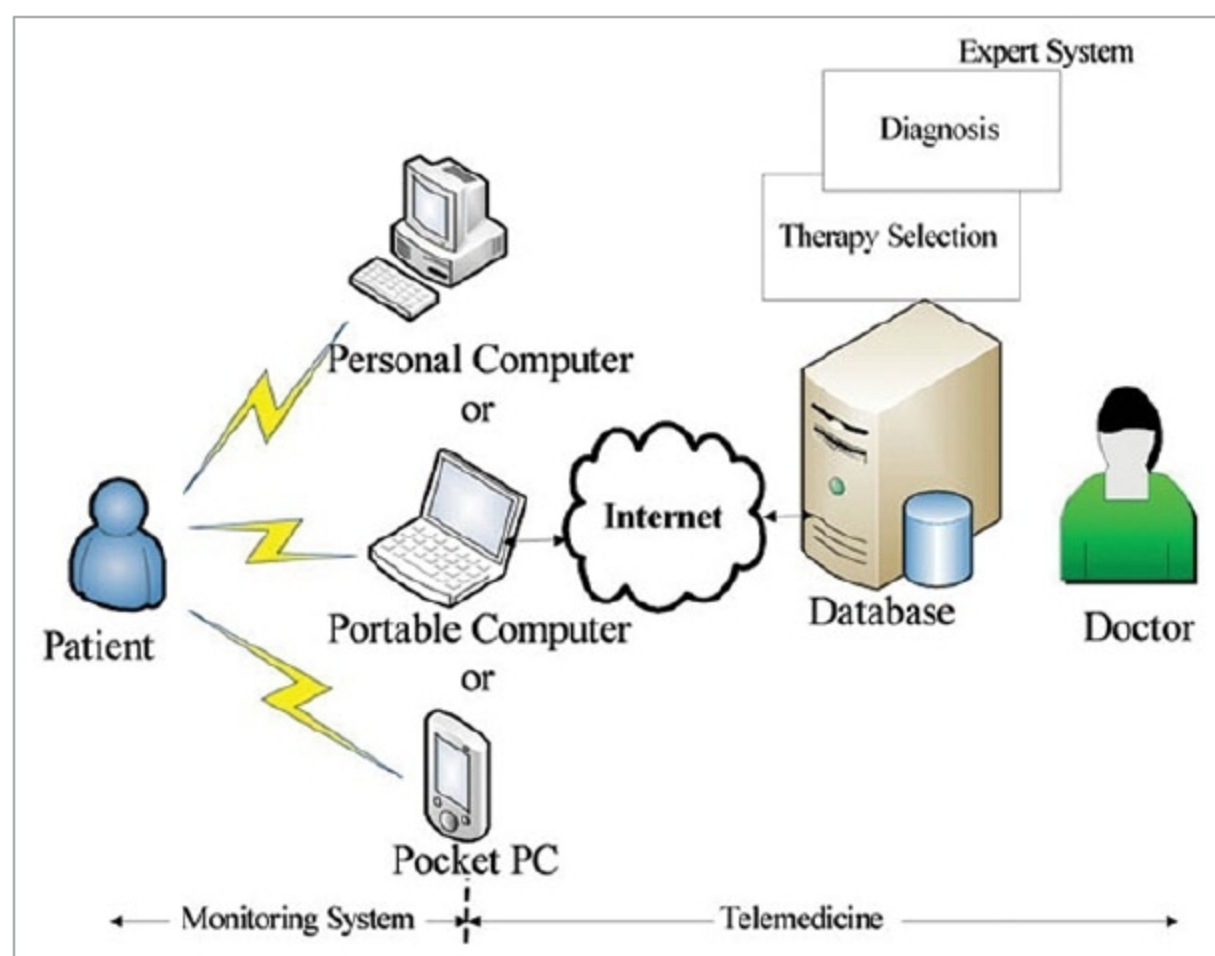




Telemedicine concept (Credit: www.scrip.org)

there are very few medical professionals to meet patients' needs. There, telemedicine can provide essential resources related to healthcare, including doctors for specific health issues in different geographic locations.

Companies are offering solutions that can help monitor patients' health remotely and automatically. Human-like *avatars* within the telemedicine interface tools ask comprehensive health questions and process the answers. This helps healthcare providers determine how to treat patients while assessing their health histories.

The ML algorithm that Google uses can diagnose diabetic retinopathy by using retinal images, which can be obtained and analysed remotely through telemedicine. ML algorithms can also detect rare genetic diseases from mere photographs of patients' faces. Patients with rare genetic disorders usually visit several doctors before figuring out the exact problem, which can now be avoided.

There are many Indian startups in the field of AI-based telemedicine, including Lybrate, mfine, Artelus, ChironX, LiveHealth, OnliDoc and Qure.ai.

- Lybrate is an online doctor-consultation platform that connects

patients with doctors and provides online consultation. Patients can also book lab tests and appointments online.

- mfine provides a virtual doctor with the ability to diagnose and triage common diseases. It refers patients with serious conditions to a hospital. Its goal is to provide healthcare to every Indian, including those in remote areas, through its virtual platform and AI software.

- OnliDoc uses AI and ML for end-to-end medical diagnosis. It has a phone application (available on Apple Store and Google Play Store) that can be used to find doctors, book appointments, and store medical records and prescriptions.

- LiveHealth is a management information system for healthcare providers. It collects samples, manages patient records, diagnoses patients and generates reports.

Digital healthcare and IT team

Digital healthcare is a broad term that includes eHealth supported by information and communication technologies, and the use of emerging and advanced technologies in the field of genomics, AI and ML, to help address the health problems faced by patients. The technologies include both hardware and software solutions,

services, telemedicine, web-based analysis, email, mobile phones and applications, text messages, wearable devices, and clinic or remote monitoring sensors.

India is among the leading countries in adoption of digital health technology. National eHealth Authority (NeHA), proposed by the Ministry of Health and Family Welfare, would be responsible for the development of an integrated health information system in India. It would also be responsible for enforcing the laws and regulations relating to the privacy and security of the patients' health and information records.

Some of the goals of digital healthcare are to reduce the cost, improve healthcare and ensure good overall patient experience. Digital transformation is an ongoing journey towards improving better healthcare that requires putting the consumer at the centre. Policy makers should put their best ideas forward to solve the healthcare crisis, with technology at the centre of discussions.

The existing technology, including information technology (IT), is falling short in many areas. For example, problems related to medical errors cause missed and delayed diagnoses, thus failing to order appropriate tests in time, or initiate follow up. The IT systems need to be designed to better manage test results and patients' related issues.

The IT department of a hospital is responsible for managing clinical software, radiology, administrative staff, patient records, admissions, medical wards, operating rooms, labour and delivery suites, and emergency departments. This is in addition to the usual tasks like providing network infrastructure, IT security, server maintenance, software deployments and so on.

The IT systems are so essential and ingrained into modern-day hospitals. Without them it can literally be a matter of life or death, if the systems stop working effectively. Traditionally, doctors and nurses are at forefront of the healthcare team. Now, with digital transformation in healthcare, we also need the best IT team and medical technologists for better healthcare. **EFY**